Emergencies and propane

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Fly the balloon first. Resolve the situation, then check the flight manual or checklist. In an emergency you may deviate from Part 91 as far as needed (91.3(b)). Your flight manual's procedure governs.

First moves (BFH Ch. 7–8)

Vent or deflation line lost	Only the burner controls descent. Stay low, land in the largest area you can, expect rebound and drag in wind; a poorer field may be worth it to avoid power lines.
Pilot light out	Two burners: fly on the other; try a relight using both. One burner: igniter or striker; else crack the blast valve slightly and light the stream, or use the backup system. Land as soon as practical.
Fuel leak or fire at a fitting	Shut off the fuel source. A small fire at a fitting: shut off, then snuff it with a gloved hand.
Thermal	Keep the envelope at least at its equilibrium temperature; don't vent to descend. It usually releases you higher.
High-wind landing	Lowest safe altitude to a large field, deflation line clear. Committed: brief passengers, fuel valves off, drain lines, pilot lights off. At 15 mph you cover 220 ft in the 10 s a burn takes to work.
Water landing	Passengers clear of the basket in case it inverts; hold on to it, it floats. A deflated envelope sinks; tanks float.
Power lines	Watch for them on every approach. If contact can't be avoided, follow your flight manual.

The checkride also asks about blast valve failure, fuel exhaustion and your emergency and survival gear (PTS Area X): answer from your flight manual.

Propane (BFH Ch. 2, App. A)

- Boils at −44 °F (butane 32 °F), but in the tank it is at the outside air temperature.
- Vapor is heavier than air and pools low; the odorant is ethyl mercaptan.
- Escaped liquid makes about 6,800 times its volume of combustible mixture.
- The fixed maximum liquid level gauge marks 80% full; the relief valve normally opens near 375 psi.

°F	psi
0	25
20	42
40	65
60	93
80	128
100	177

Cold mornings: add nitrogen through a regulator if the flight manual allows; vapor-feed pilot systems may not tolerate it. Cancelled, with a rise of more than 30° coming? Bleed the pressure through the fixed liquid level gauge.

Refueling: open air, never from inside the basket or a closed trailer or van · no smoking · strikers disabled · chase vehicle off · gloves. When the spit valve spits: main tank valve, then spit valve, then supply line, then bleed the lines.

Sources: 14 CFR 91.3, FAA-H-8083-11B Ch. 2 (fuel tanks, propane primer), FAA-H-8083-11B Ch. 3 (cold weather operations), FAA-H-8083-11B Ch. 7 (in-flight emergencies), FAA-H-8083-11B Ch. 8 (thermals, high-wind and water landings, refueling), FAA-H-8083-11B Appendix A, FAA-S-8081-18A Areas X and XI. Original card, not an FAA document. Study aid, not for navigation.